



Faculty of Computer Science

Data Science & Business
Analytics

Moscow 2025

Implementing Machine Learning for Optimizing Debit-Product Marketing

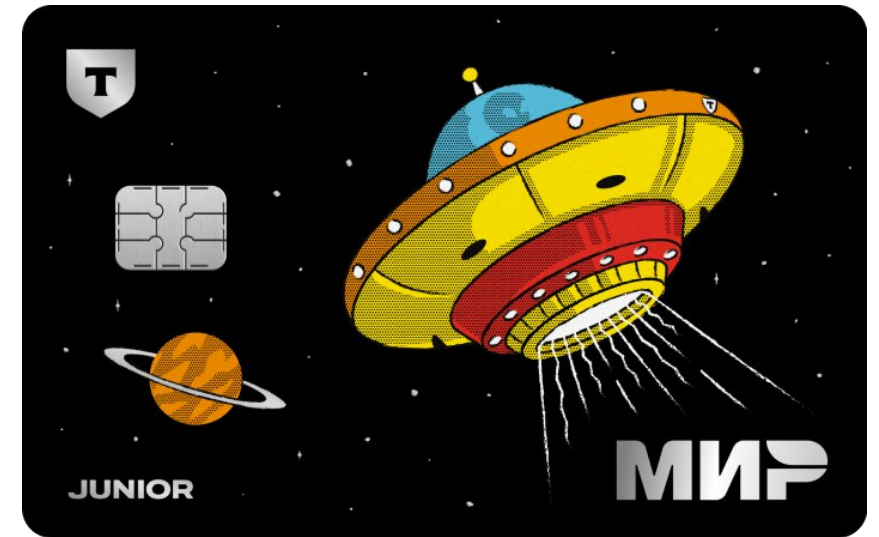
Performed by a student of DSBA211 group
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Campaign Analytics Group: segments,
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Motivation & Market Gap

Challenge	Impact
Historical conversion rate $\approx 5\%$	High media spend, low ROI
Broad targeting to entire base	Customer fatigue $\rightarrow$ spam complaints, unsubscribes $\rightarrow$ block access to other product offers
No channel optimisation	Bank wastes money sending expensive SMS to users who might open via a free banner instead
Competitors use AI targeting	Risk of falling behind in relevance and cost efficiency



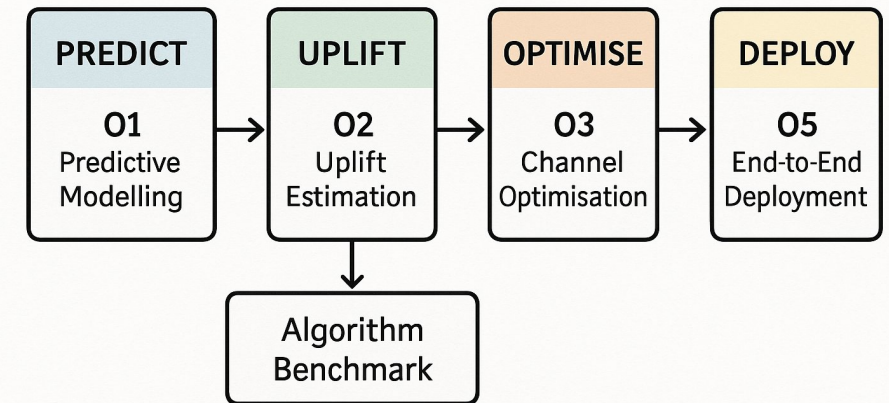


Aim and Objectives

Aim: Create and productionise ML-driven marketing system that raises Junior-Card incremental conversions by 1.5 – 2 times while keeping spend flat.

Objectives:

Objective	Deliverable
O1 Predictive Modelling	Robust model that ranks every customer's baseline likelihood to open a Junior Card (handles severe class imbalance).
O2 Uplift Estimation	Quantify the causal lift of each communication using Two-Model and Class-Transformation frameworks.
O3 Channel Optimization	Bayesian layer that selects e-mail, push, banner, chat, or SMS to maximize individual uplift.
O4 Algorithm Benchmark	Benchmark all feasible classical machine-learning algorithms for this task, evaluate accuracy and uplift.
O5 End-to-End Deployment	Integrate scorer into T-Bank's A/B platform; analyze developed model performance on real users; make architecture reusable for other products.





Why ML + Uplift & Research Questions

Why Machine Learning?

- Processes 480 M touch-points, captures non-linear patterns.
- Adapts to behavioural drift across five channels.

Why Uplift Modelling?

$$\Delta(x) = P(Y=1 | T=1, x) - P(Y=1 | T=0, x)$$

- Measures causal effect of a message, not just likelihood.
- Enables budget to focus on true persuadables.

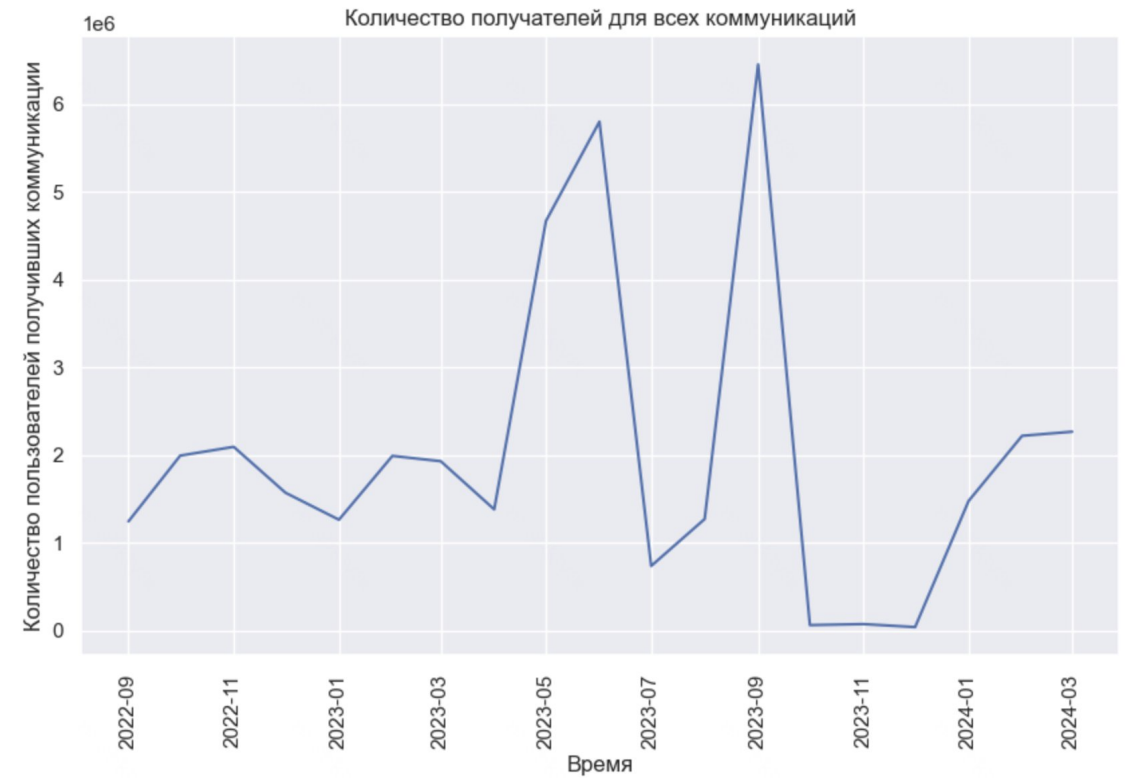
Research Questions

- Which algorithm works best for predicting Junior Card adoption?
- Which uplift method gives more reliable results — Two-Model or Class-Transformation?
- Does adding Bayesian channel scores improve performance in real campaigns?



Data Landscape

Data Source	Rows	Key Fields
Core registry	50 M	age, gender, region, device id
Junior-Card ledger	~3 M	sign-up date, referral
Communication log	480 M	channel, send/seen/click time-stamps
App telemetry	11 B	sessions, taps, dwell-time





Key Exploratory Insights

- Spike $\neq$ success — traffic peaks did not raise per-channel utilization (Fig 1–2).
- Push > Banner > E-mail — raw uplift differs by channel; opportunity for routing.
- Seasonality mild — no month shows >0.2 pp deviation once volume is normalized.

Action: encode empirical success as Bayesian priors.



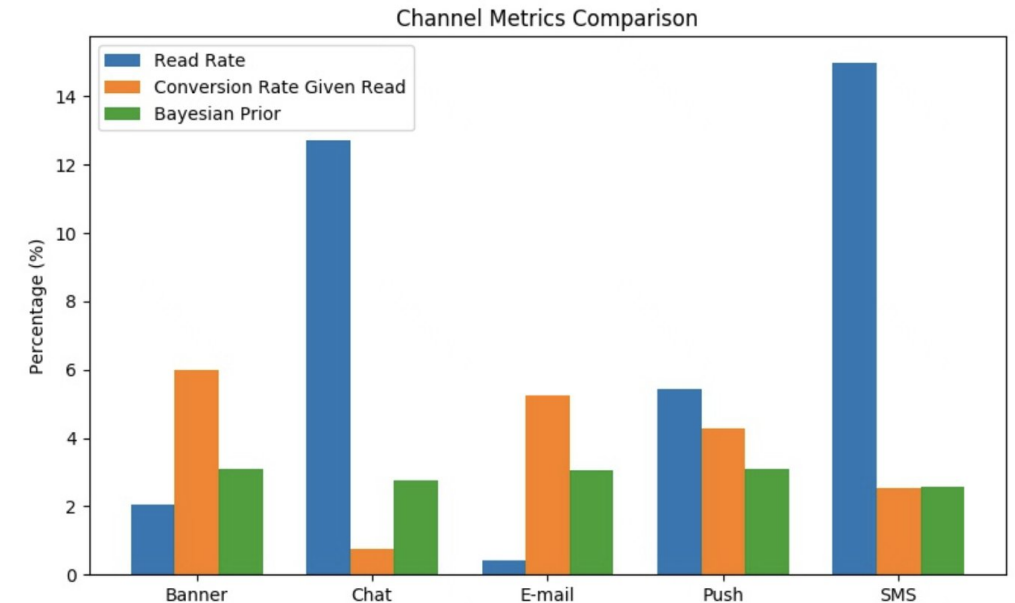
Bayesian Channel Priors

$$\pi_c = \frac{N_c^{\text{card}}}{N_c^{\text{contact}}}, \quad \tilde{\pi}_c = \frac{N_c^{\text{card}} + \alpha}{N_c^{\text{contact}} + 2\alpha}, \quad \alpha = 10$$

Posterior expectation refreshed after every A/B batch

$$\tilde{\pi}_{i,c}^{(t)} = \frac{\alpha_c^{(t-1)} + y_{c,t}}{\alpha_c^{(t-1)} + \beta_c^{(t-1)} + n_{c,t}}$$

- $\alpha_c^{(t-1)}$ - prior “wins” for channel **c** before batch **t**
- $\beta_c^{(t-1)}$ - prior “losses” for channel **c** before batch **t**.
- $y_{c,t}$ - actual wins in channel **c** during batch **t**.
- $n_{c,t}$ - total contacts sent via channel **c** in batch **t**.



Channel	F1 (OOS)	ROC-AUC (OOS)
Push	0.666	0.690
Banner	0.655	0.705
E-mail	0.540	0.661
SMS	0.602	0.650



Feature Engineering & Selection

Blocks & examples:

- Demographics → age, gender, marital status
- Transactions → mean ticket, max balance, MCC entropy
- Portfolio → debit-share %, credit count
- Engagement → min/avg/max daily sessions (90 d)
- Social graph → share of friends already with Junior Card
- Channel history → push delivered, e-mail opened, banner dwell
- Bayesian priors $\tilde{\pi}_c \rightarrow 5$ smoothed probabilities

Selection pipeline:

- Dummy probes (95 th percentile filter)
- Boruta-RF ($p < 0.05$ retain)

$$\Pr(I_j \leq I_j^{\text{shadow}}) < 0.05.$$

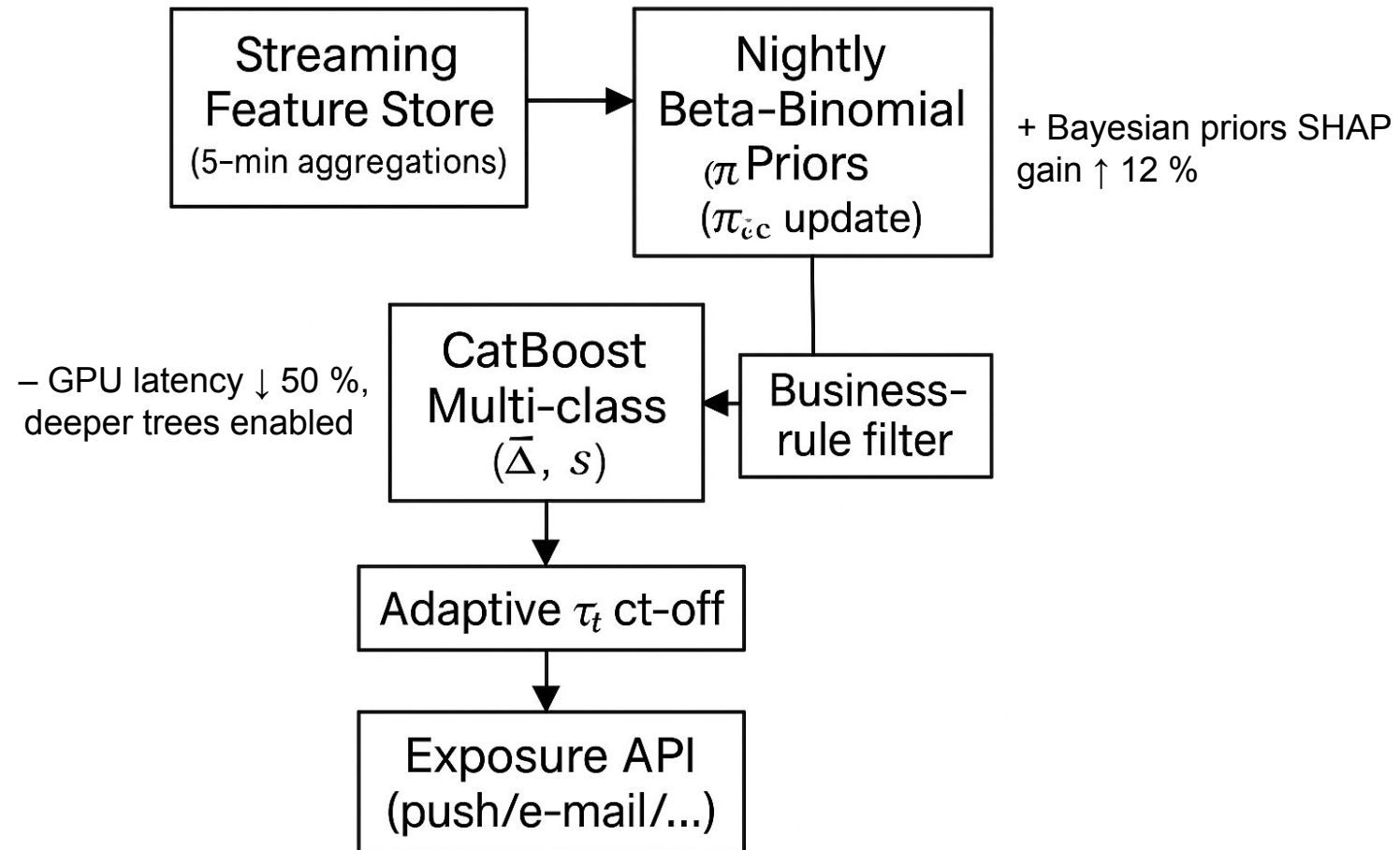
- CatBoost SHAP sweep → 87 final features



Baseline Benchmark

Metric	Baseline (old business rules)	Logistic Reg.	Naïve Bayes	k-NN	Decision Tree	Random Forest	Extra Trees	SVM (RBF)	XGBoost	LightGBM	CatBoost
CV ROC-AUC	0.55	0.728	0.687	0.701	0.745	0.782	0.791	0.764	0.804	0.815	0.823
CV F1	0,42	0.69	0.63	0.66	0.68	0.74	0.75	0.71	0.76	0.77	0.78
Hold-out ROC-AUC	0.53	0.703	0.671	0.692	0.721	0.745	0.753	0.713	0.769	0.773	0.777
F1 Top-10 %	0.052	0.112	0.085	0.098	0.104	0.161	0.168	0.146	0.174	0.181	0.191
Train-time	-	40 s	15 s	8 min	20 s	90 min	60 min	185 min	110 min	55 min	120 min

Proposed Pipeline



Uplift Modelling Framework

$$\Delta(x) = P(Y = 1 \mid X = x, T = 1) - P(Y = 1 \mid X = x, T = 0)$$

Version 1: $\hat{\Delta}_1(x) = \hat{p}_1(x) - \hat{p}_0(x)$

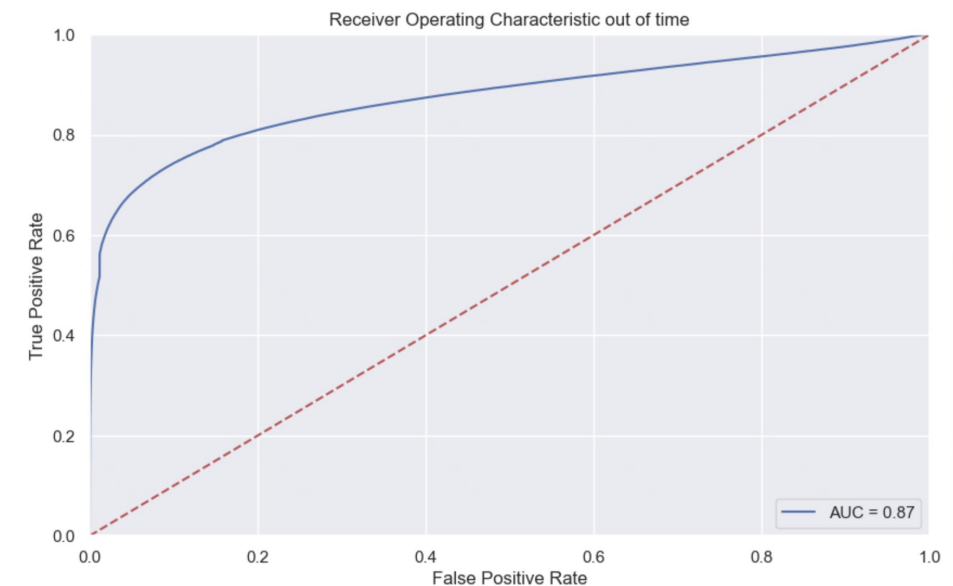
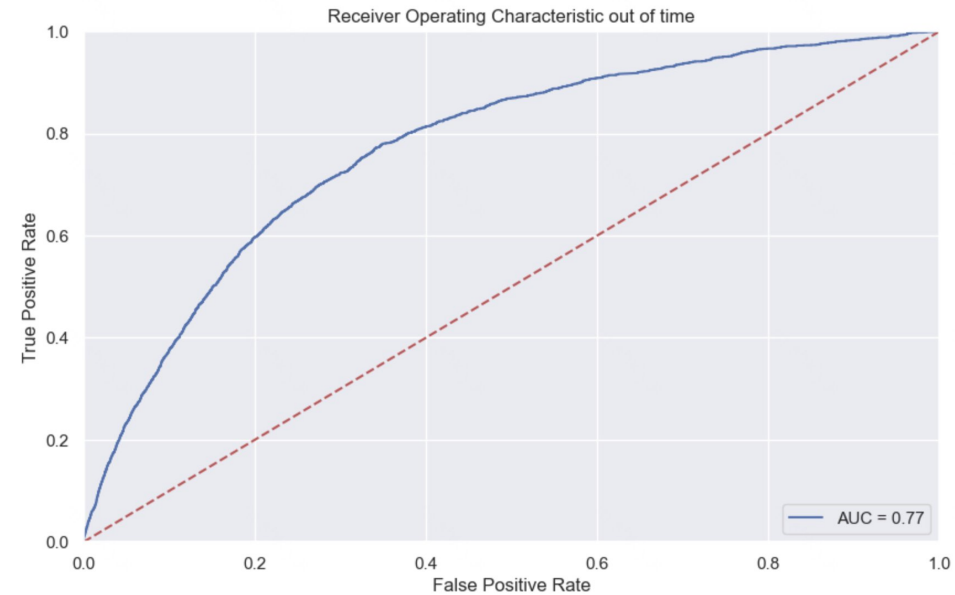
Version 2 (Radcliffe–Gérardy class-transformation)

$$z = \begin{cases} 1 & (t=1, y=1) \\ 2 & (t=1, y=0) \\ 3 & (t=0, y=1) \\ 4 & (t=0, y=0) \end{cases}, \quad \hat{\Delta}_2(x) = [p(z=1 \mid x) - p(z=2 \mid x)] - [p(z=3 \mid x) - p(z=4 \mid x)]$$



Classification Results

Split	ROC AUC	Accuracy	F1
Train (V1)	0.936	0.862	0.823
Train (V2)	0.938	0.860	0.862
Validation (V1)	0.895	0.811	0.757
Validation (V2)	0.898	0.808	0.810
Test-1 (out-of-time) V1	0.884	0.786	0.789
Test-1 (out-of-time) V2	0.874	0.796	0.767
Test-2 (out-of-sample) V1	0.763	0.796	0.011
Test-2 (out-of-sample) V2	0.892	0.816	0.818
Test-3 (OOS + OOT) V1	0.763	0.796	0.011
Test-3 (OOS + OOT) V2	0.891	0.816	0.818





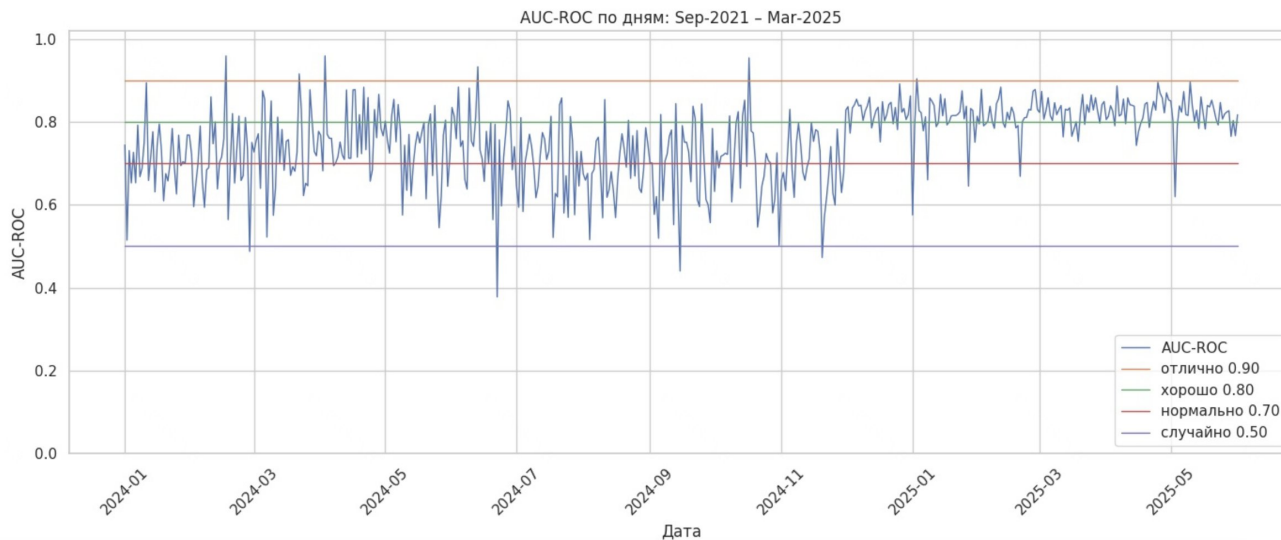
Uplift Results

Metric (Test-2)	V1	V2
uplift % (overall)	0.45 pp	0.60 pp
uplift % (by-group)	0.64 pp	0.80 pp
weighted-average uplift	0.02 pp	0.08 pp
AUUC	0.0017	0.0033

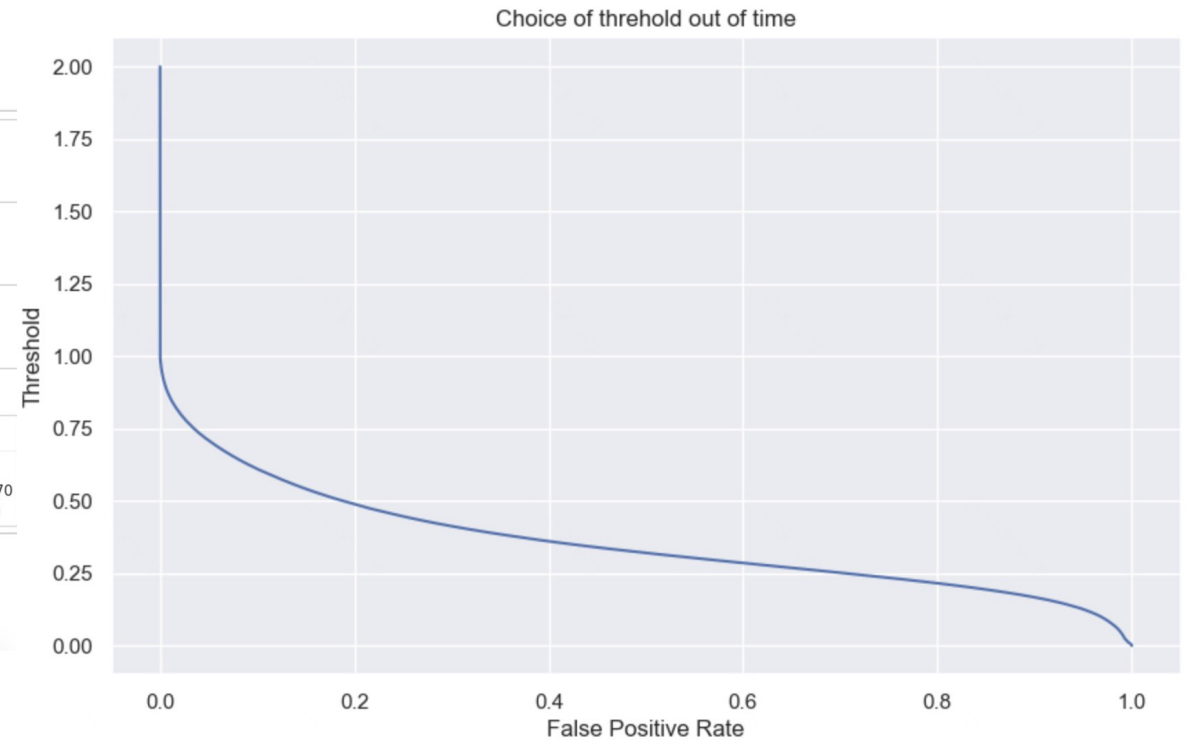


Model Stability & Decision Threshold

Daily AUC-ROC Stability: V2 spends most days > 0.80 (“good”) and frequently touches $0.85+$, while V1 exhibited long dips below 0.70



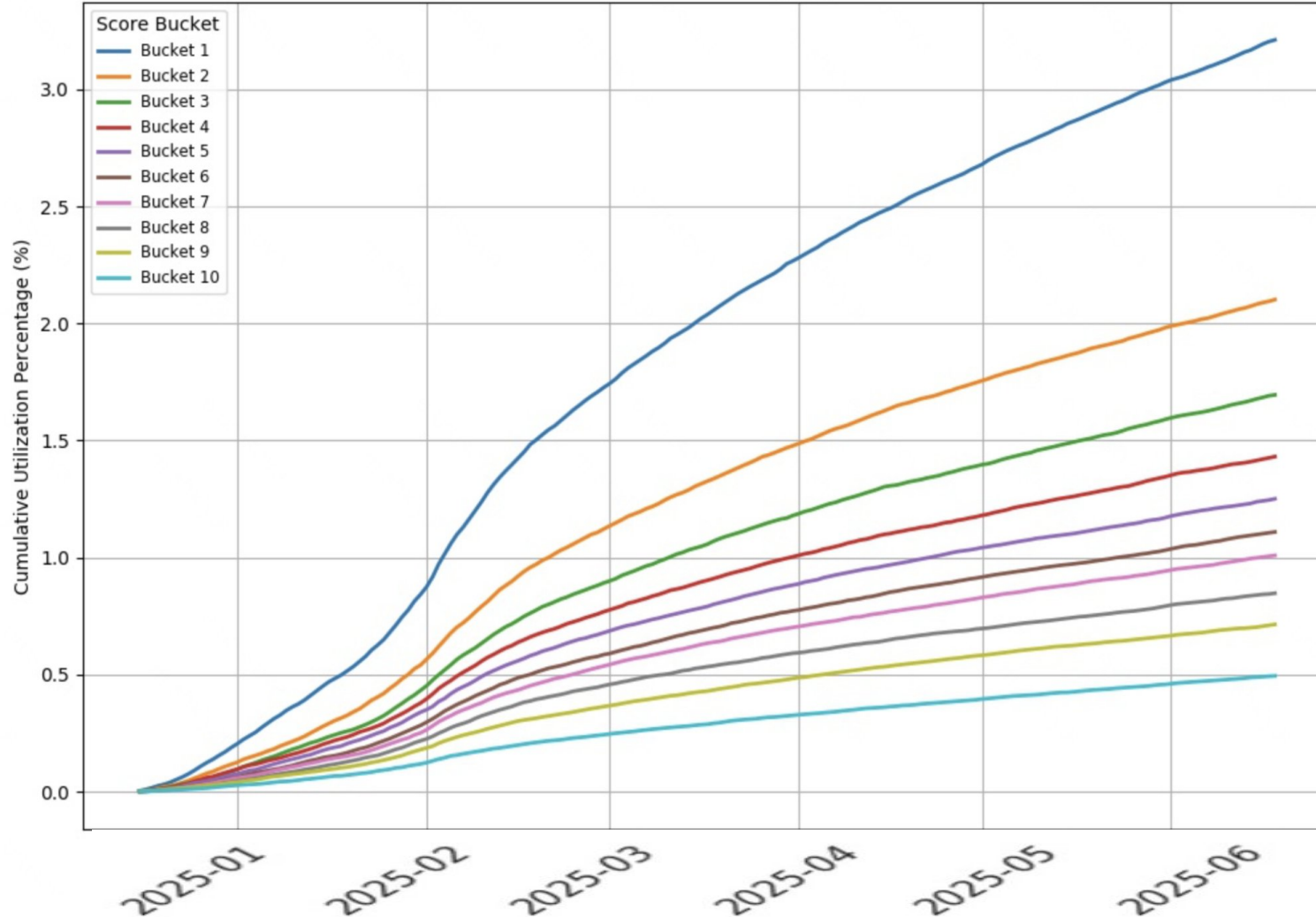
FPR–Threshold Curve: A smooth, single-knee profile $\Rightarrow$ one stable cut-off; optimal threshold ≈ 0.48 keeps $FPR < 0.25$ with minimal variance





Score-to-Outcome Ordering

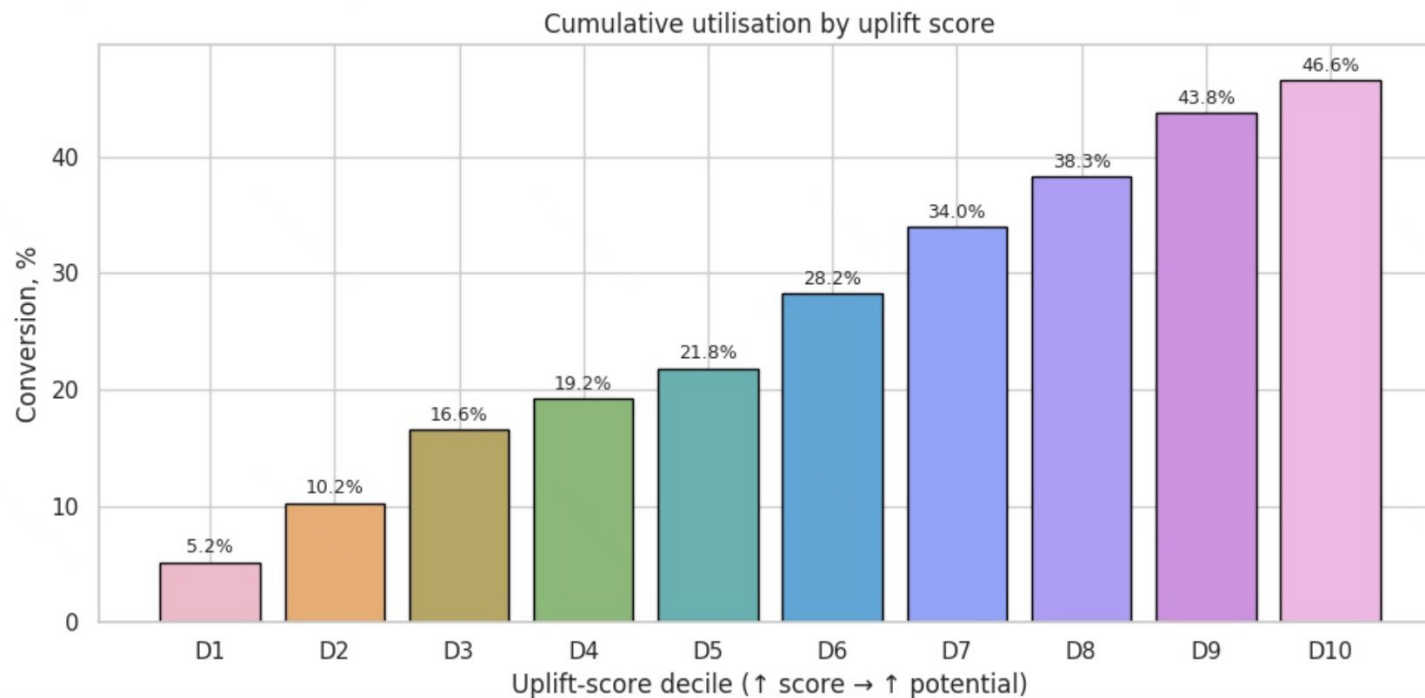
Cumulative Utilization Percentage by Score Bucket



- Perfect monotonic fan-out: top-decile curve stays on top throughout the six-month horizon.
- Gap between Decile 1 and Decile 10 widens from 0 → 3 pp, proving that the static onboarding score predicts long-run behaviour.
- Median time to utilisation in the highest bucket is less than half that of the overall population.



Business Impact



- **Conversion lift:**
contacted CVR 5.2 % → 10.3 % (+≈2 ×)
- **Uplift rise:** as the score rises,
the uplift between successive groups
grows monotonically.
- **Cost efficiency:** blended CPA ↓ 14 %
- **Smarter channel mix:**
push + banner ↑, e-mail volume ↓ 18 %
- **Customer experience:**
NPS ↑, opt-outs & spam complaints ↓
- **Operational leverage:**
product-agnostic, GPU-light pipeline



Research Limitations

Short horizon – eight-day window captures only early uptake

Nightly priors – intra-day shocks currently invisible

External drift – holidays & policy changes may shift baseline

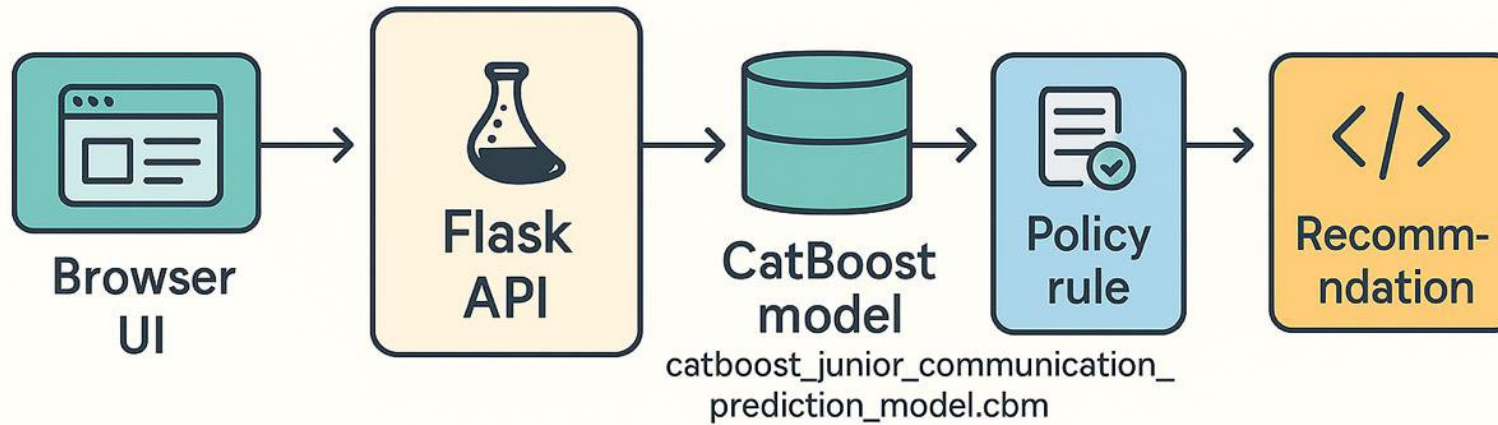
Unobserved confounders – uplift still assumes ignorability

Privacy constraints – strict PII rules limit new feature types

Channel ceilings – fixed caps on push/SMS mean we can't always shift spend to the statistically best channel in real time

Decile rigidity – using a strict top-10 % cutoff may skip users just below the threshold who still have positive uplift

Application Prototype – “Junior-Card Marketing Helper”



- Web UI (10 top features)
- RESTful endpoint
- Feature mapping → model inference
- Uplift & propensity calculation
- Policy rule & verdict



Demonstration

Junior-Card Marketing Helper

Primary Channel	Age	Gender	Letters 3 mo	Debit Apps Σ
push	32	Female	120	1
			delivered e-mails	applications filed
Debit-Product Ratio	Min Daily Sessions	Avg Daily Sessions		Max Daily Sessions
0.65	0,5	3,2		7,8
Predict uplift				

Prediction Result

Recommendation: **Send the communication.**

Feature	Value
channel	push
age	32
letters	120
debit_apps	1
debit_ratio	0.65
min_act	0.5
avg_act	3.2
max_act	7.8
gender	Female

[Demonstration Link](#)



Conclusions

End-to-end ML stack:

feature store → Bayesian prior → CatBoost uplift scorer

Best-in-class quality:

AUC 0.87 OOT, uplift@10 % +0.60 pp

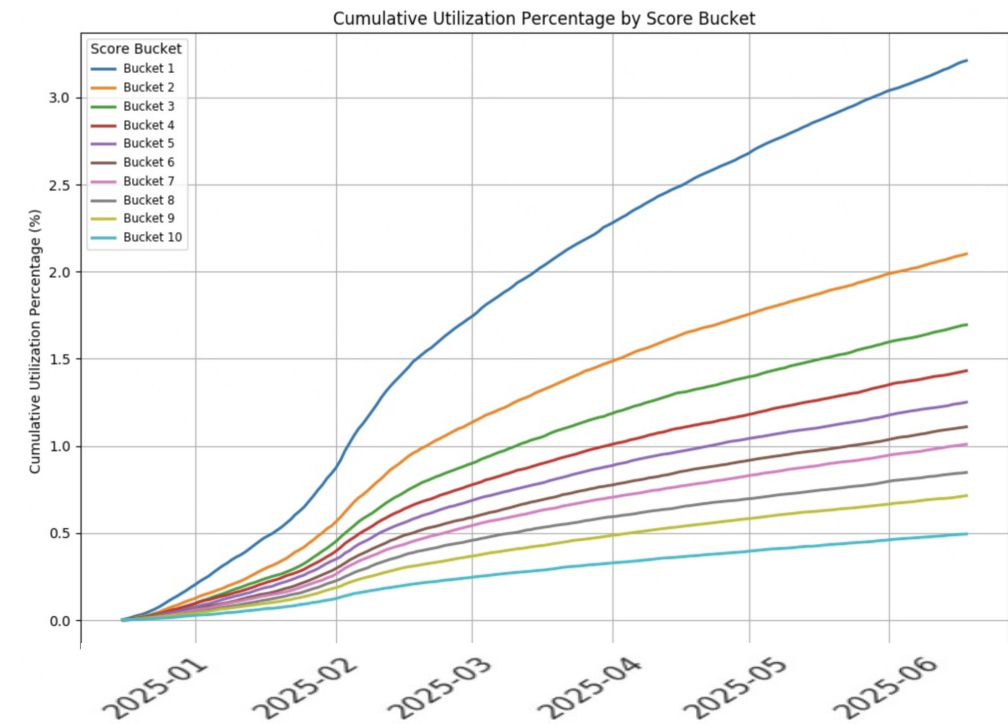
Business lift:

contacted CVR 5.2 % → 10.3 %, CPA ↓ 14 %,
e-mail load ↓ 18 %

Reusable architecture:

product-agnostic, GPU-light, easy to clone

Validated ordering: utilization curves separate cleanly



Future work

Stream scores – move feature store to Kafka for sub-5 min refresh

Sharper uplift – causal loss + meta-learner ensemble

Adaptive channels – RL bandit to react to fatigue in-flight

Multi-product goals – optimize debit + credit + wealth under one budget

Always-on quality – drift detector → auto-retrain





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Thank you for your attention!

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